



EDLANX PRIVATE LIMITED

CYBER SECURITY CURRICULUM

1. Introduction to Cybersecurity

1.1.1. Overview of Cybersecurity

1.1.2. Threat Landscape and Attack Vectors

1.1.3. Cybersecurity Frameworks and Standards

2. Operating Systems and Networking Fundamentals

2.1.1. Introduction to Operating Systems

2.1.2. Networking Concepts and Protocols

2.1.3. Network Security Fundamentals

3. Cryptography and Steganography

3.1.1. Cryptography Basics (Symmetric and Asymmetric Encryption, Hashing, Digital Signatures)

3.1.2. Cryptographic Attacks and Countermeasures

3.1.3. Steganography and Steganalysis

4. Ethical Hacking

4.1.1. Introduction to Ethical Hacking

4.1.2. Reconnaissance Techniques

4.1.3. Scanning and Enumeration

4.1.4. Vulnerability Assessment and Penetration Testing

4.1.5. Exploitation Techniques and Post-Exploitation

5. Web Application Security

5.1.1. Common Web Application Vulnerabilities (SQL Injection, Cross-Site Scripting, CSRF)

5.1.2. Web Application Penetration Testing

5.1.3. Secure Coding Practices for Web Applications

6. Network Security

6.1.1. Network Access Control and Authentication

6.1.2. Firewall and Intrusion Detection Systems

6.1.3. Virtual Private Networks (VPNs) and Secure Sockets Layer (SSL)

7. Cloud Security

7.1.1. Cloud Computing Fundamentals

7.1.2. Cloud Security Architecture and Models

7.1.3. Cloud Security Threats and Countermeasures

8. Mobile Security

8.1.1. Mobile Security Fundamentals

8.1.2. Android and iOS Security Architecture

8.1.3. Mobile Application Security Testing

9. Social Engineering

9.1.1. Introduction to Social Engineering

9.1.2. Social Engineering Techniques (Phishing, Vishing, Pretexting)

9.1.3. Social Engineering Countermeasures

10. Incident Response and Digital Forensics

10.1.1. Incident Response and Handling

10.1.2. Digital Forensics Fundamentals

10.1.3. Forensic Analysis of Digital Evidence

11. Cybersecurity Governance and Compliance

11.1.1. Cybersecurity Governance Frameworks and Standards

11.1.2. Compliance Requirements and Regulations (GDPR, HIPAA, PCI DSS)

12. Ethics in Cybersecurity

12.1.1. Professional Ethics and Conduct

12.1.2. Legal and Regulatory Issues

12.1.3. Ethical Hacking and Responsible Disclosure

13. Advanced Topics in Cybersecurity

13.1.1. Advanced Persistent Threats (APTs)

13.1.2. Internet of Things (IoT) Security

13.1.3. Artificial Intelligence (AI) in Cybersecurity

13.1.4. Blockchain Security

14. Projects:

14.1.1. Vulnerability Assessment and Penetration Testing: Develop a comprehensive penetration testing plan for a web application or network, including vulnerability scanning, exploitation of identified vulnerabilities, and remediation recommendations.

14.1.2. Social Engineering Awareness Training: Develop a social engineering awareness training program for employees, including simulation exercises and education on common social engineering techniques.